BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: V** **Roll No.: TYIT 11** **Date : 17 /07/2026**

Course Name: Big Data Analytics **Page no:**

Practical Number:- 7

Aim :- Implement Decision tree classification techniques.

Code:

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, export_text, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, accuracy_score
import matplotlib.pyplot as plt

# 1.Load dataset
iris=load_iris()
X=iris.data#features
y=iris.target #target classes

# 2. Split into training/testing
X_train, X_test, y_train, y_test=train_test_split(X,y,test_size=0.3,random_state=42)

#3. train decision tree
model=DecisionTreeClassifier(criterion='entropy', max_depth=3, random_state=0)
model.fit(X_train, y_train)

#4. predict ontest data
y_pred=model.predict(X_test)

#5. Evaluate the model
print("Classification report:\n",classification_report(y_test,y_pred))
print("Accuracy :", accuracy_score(y_test, y_pred))

#6. display the tree as text
print("\nDecision Tree Rules:\n")
print(export_text(model, feature_names=iris.feature_names))

#7. plot the decision tree(optional)
plt.figure(figsize=(12,8))

plot_tree(model, feature_names=iris.feature_names, class_names=iris.target_names, filled=True)

plt.title("Decision Tree Visualization Ravit TYIT_11")
plt.show()
```

Teacher's Signature

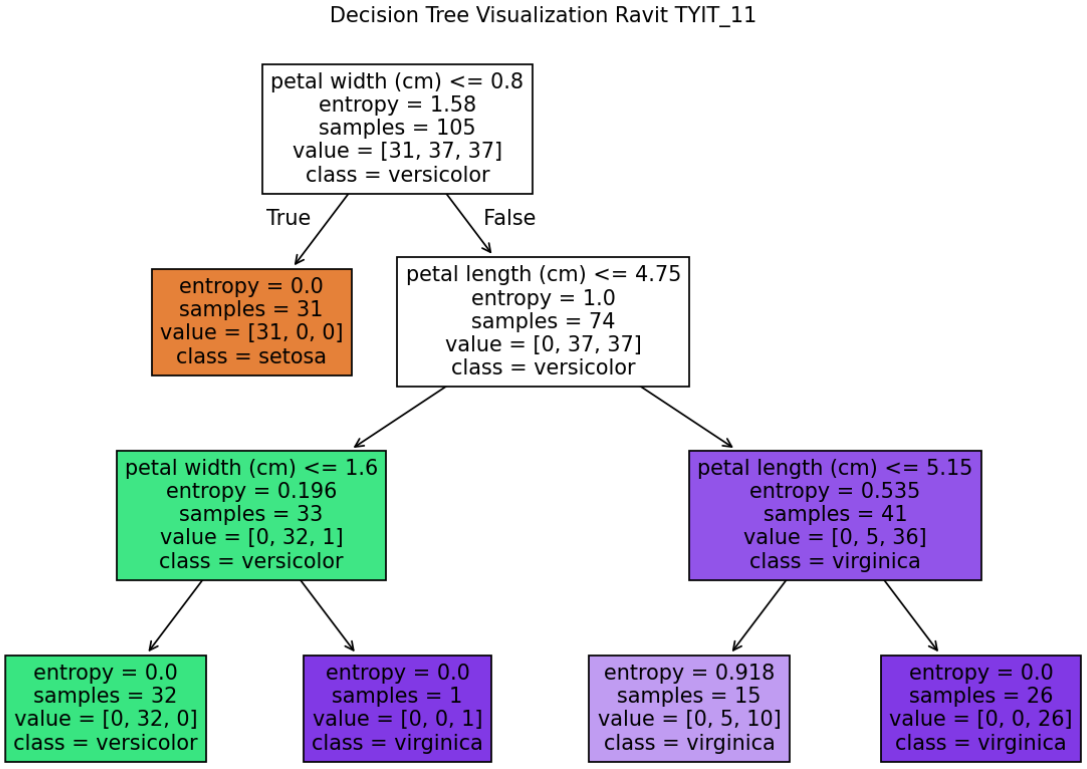
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Output :




```
============== RESTART: C:/Users/Admin/OneDrive/Desktop/TYIT11.py
Classification report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      0.92      0.96        13
           2       0.93      1.00      0.96        13

    accuracy                           0.98        45
   macro avg       0.98      0.97      0.97        45
weighted avg       0.98      0.98      0.98        45

Accuracy : 0.9777777777777777

Decision Tree Rules:

|--- petal width (cm) <= 0.80
|   |--- class: 0
|--- petal width (cm) >  0.80
|   |--- petal length (cm) <= 4.75
|   |   |--- petal width (cm) <= 1.60
|   |   |   |--- class: 1
|   |   |--- petal width (cm) >  1.60
|   |   |   |--- class: 2
|   |--- petal length (cm) >  4.75
|   |   |--- petal length (cm) <= 5.15
|   |   |   |--- class: 2
|   |   |--- petal length (cm) >  5.15
|   |   |   |--- class: 2
```

Teacher's Signature